

Project Name: Impact of Commuting and Housing on Enrollment

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Problem:

A [tableau dashboard](#) created by UIC's Office of Institutional Research showed the number of students from each Illinois county enrolled in each university under the University of Illinois system in Fall 2023. Most students going to a UI university. The number of students from each UI school in a county was not evenly distributed. I believe factors such as housing and commutation services play a huge role in the decision a student makes on where to enroll. available to students going to that university. Analyzing this information will be helpful in identifying areas where the university may be missing opportunities to attract students and offer better services and infrastructure to accommodate students from counties that have fewer students from their universities.

Data:

I plan to use the following data for my project:

1. UIC Enrollment by County: This data is available through an interactive Tableau map, showing the number of students from each Illinois county enrolled in University of Illinois schools for Fall 2023. I will need to manually copy the data into a spreadsheet, which will take around 30 minutes. The dataset will contain fields for county, university, and number of students.
2. Distances from Each County to UI Schools: This data will be collected using Google Maps to determine the driving distance between each Illinois county and the University of Illinois schools. The process is straightforward and doesn't require a lot of time. The dataset will include fields for county, university, and distance (in miles).
3. Housing Costs at Each University: This data is available on each university's website. I will compile the average housing costs for each campus and organize them into a spreadsheet. The dataset will have fields for university and average housing costs.

Solution:

I will approach this problem by first collecting and cleaning all the data required for the project. Then, I will look for patterns in how distance from the university affects a student's choice to attend. I will also look for relationships between housing costs and the number of students from a county who live past a commuting distance. The scope of this project is limited to UI schools and publicly available data from their websites. To analyze the information, I will create visualizations such as distance vs. enrollment graphs and a heat map of enrollment by county. I expect the end result to show a strong correlation between proximity to the university, housing costs, and enrollment numbers. I envision the system to be static. By the time of the progress report, I aim to have collected and cleaned all the necessary data and begun exploring visualizations to identify correlations.